INTRODUCTION TO HYBRID AND ELECTRIC VEHICLES

Subject Code : 23A0360Se

UNIT III

Hybrid and Electric Drive Trains

I. Introduction to Hybrid Electric Vehicles

A **Hybrid Electric Vehicle (HEV)** combines two or more energy sources / power sources - normally an **internal combustion engine (ICE)** and an **electric motor with battery** - to drive the vehicle. The engine can be run at its most efficient point and the electric path recovers braking energy.

Concept of hybridisation

- **Micro hybrid** - stop-start only. **Mild hybrid** - small motor assists engine (BAS, 48 V). **Full hybrid** - can run in pure electric mode (Prius). **Plug-in hybrid** - large battery, grid charging.
- Fuel economy gain: 10-15 % (mild) to 30-50 % (full hybrid) in city driving.

History of hybrid vehicles

- 1830s-1890s: first electric carriages; 1900: **Lohner-Porsche** series hybrid with in-wheel hub motors.

- 1900-1920: petrol cars (Ford Model T, electric starter 1912) displaced EVs because of cheap oil and long range.
- 1970s: oil crises renewed interest; 1990s: California ZEV rules. 1997: **Toyota Prius** (Japan) - first mass-produced hybrid. 1999: **Honda Insight** in USA. 2000s: Ford Escape, Lexus hybrids. 2010: **Chevrolet Volt** (plug-in), **Nissan Leaf** (BEV). Now: Tesla, BYD, Tata Nexon EV etc.

Social importance

- Reduces dependence on imported petroleum (energy security) and fuel bills.
- Lowers CO_2, NO_x, HC and particulate emissions - better urban air quality and less global warming.
- Less noise; creates new technology industry and jobs; supports renewable energy via grid charging.

Impact of modern drive trains on energy supplies

- Transport uses about 25-30 % of world energy and is 95 % oil-based. Higher efficiency drive trains reduce oil demand; each 1 % fuel saving across the fleet is a large barrel-per-day saving.
- PHEV / EV shift part of transport energy to the **electric grid**: increases electricity demand but allows use of coal, gas, nuclear, solar and wind; **well-to-wheel efficiency** of

EV (about 25-35 %) is better than ICE (about 12-20 %).

- Smart charging and vehicle-to-grid (V2G) can support grid stability; battery material (lithium, cobalt) supply and recycling become new issues.

2. Hybrid Traction and Electric Traction

Aspect	Hybrid traction	Electric traction
Energy source	ICE (fuel) + battery / motor	Battery / fuel cell / ultracap only
Emission	Low, some tail-pipe	Zero tail-pipe
Range	Long (fuel tank)	Limited by battery
Refuelling	Petrol + optional plug	Charging / H_2 filling
Complexity	High (2 power plants)	Simple drive train
Examples	Prius, Insight, Volt	Leaf, EVI, Tata Nexon EV

Hybrid traction uses ICE and electric machine in series, parallel or combined arrangement. **Electric traction** uses only electric machines with energy from battery, fuel cell or supercapacitor; also used in electric trains and trams.

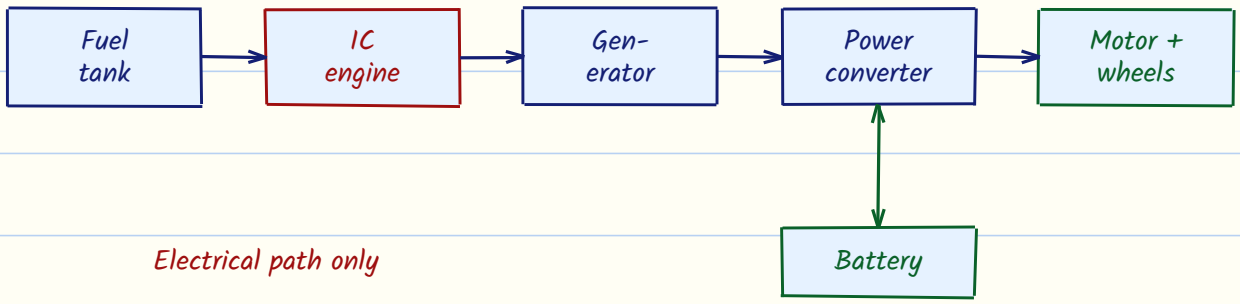
3. Hybrid and Electric Drive Train Topologies

(a) Series hybrid

The ICE drives a **generator**; electric power goes to the battery and traction motor. Only the motor is coupled to the wheels. The engine speed is independent of vehicle speed, so

it can run at optimum efficiency.

Series hybrid: engine NOT mechanically connected to wheels

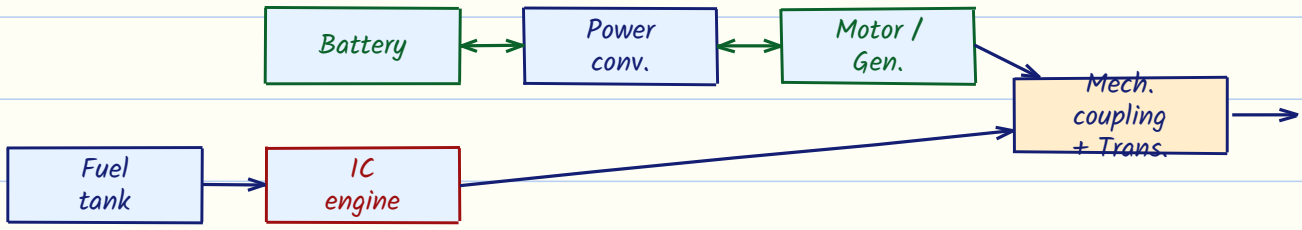


- + Simple control, engine at best efficiency point, no clutch or gear box, flexible layout.
- - Energy is converted twice (mech -> elec -> mech), so lower efficiency at highway speed; needs three machines of full rating (engine, generator, motor). Suitable for city buses, locomotives, range extenders (Volt in extended mode).

(b) Parallel hybrid

Both ICE and electric motor are connected to the transmission through a **mechanical coupling**; they can drive wheels together or separately. The motor also acts as generator / starter.

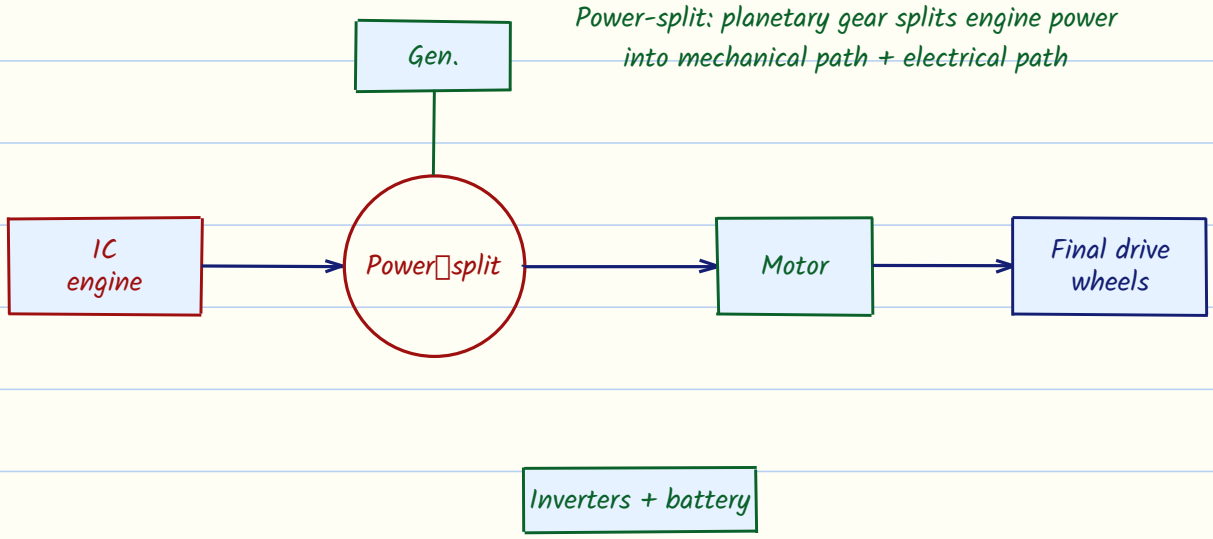
Parallel hybrid: engine and motor both drive the wheels



- + Smaller motor and battery, only one electric machine, high efficiency at highway, no energy conversion when engine drives.
- - Complicated mechanical coupling and control, engine speed linked to wheels so not always at best point. Ex: Honda Insight (IMA), mild hybrids.

(c) Series-parallel (power-split) hybrid

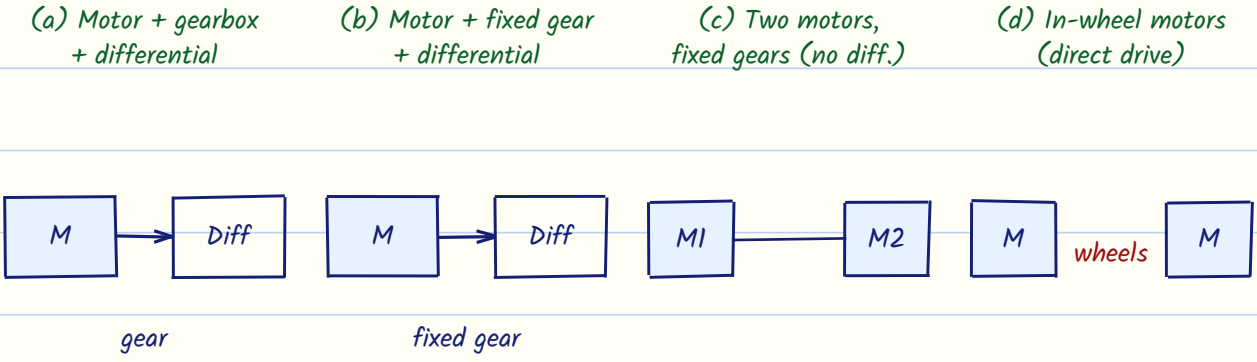
A **planetary gear (power split device)** divides engine power into a mechanical path (to wheels) and an electrical path (generator -> motor). Gives the advantages of both: engine speed is decoupled from wheel speed (like CVT) and direct mechanical drive. Ex: **Toyota Prius**.



(d) Complex hybrid

Similar to series-parallel but the generator and motor can switch roles using more power electronics (bi-directional). More flexible but costly. Ex: Lexus RX400h (four-wheel drive).

Electric (BEV) drive train topologies



- (a) is the conventional layout; (b) removes gearbox - single-speed reduction; (c) two motors give electronic differential; (d) **in-wheel / hub motors** remove all

transmission but raise unsprung mass.

4. Power Flow Control and Energy Efficiency Analysis

Power flow control means deciding, at each instant, how much power comes from engine, motor/generator and battery so that the **demand** at the wheel is met with minimum fuel and the battery SOC is kept within limits.

Operating modes of a series hybrid

- **Pure electric (low speed)** - battery -> motor -> wheels, engine off.
- **Engine alone (cruise)** - engine -> generator -> motor -> wheels, battery idle.
- **Hybrid (acceleration)** - engine-generator and battery both supply the motor.
- **Battery charging** - engine -> generator -> battery (vehicle stopped or light load).
- **Regenerative braking** - wheels -> motor (as generator) -> battery.
- **Combined**: charging while driving; **stand still** - engine off.

Operating modes of a parallel hybrid

- Motor only (start / low load); engine only (highway); engine + motor (peak power); engine drives vehicle and charges battery through motor-generator; regenerative

braking; stop-start.

Power balance equations

$$P_{wheel} = P_{engine}\,\eta_{m1} + P_{motor}\,\eta_{m2}$$
$$P_{batt} = (P_{motor} / \eta_m) - (P_{gen}\,\eta_g)$$

Supervisory (energy management) controller chooses the split by rules (thermostat / power-follower) or by optimisation to keep ICE in its high-efficiency zone and SOC between 40 % and 80 %.

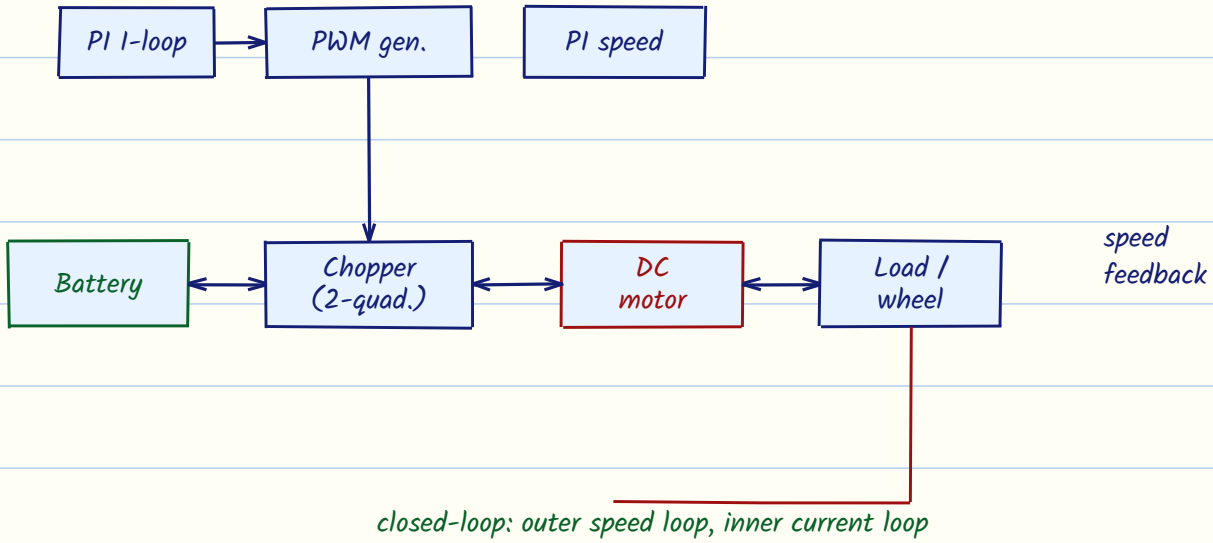
Energy efficiency analysis

- Efficiency of a path = product of component efficiencies. Series path: $\eta_{ICE}\,\eta_{gen}\,\eta_{conv}\,\eta_{batt}\,\eta_{motor}\,\eta_{trans}$ (low); parallel direct path: $\eta_{ICE}\,\eta_{trans}$ (high).
- Fuel saving comes from: (i) engine downsizing, (ii) running engine near best brake specific fuel consumption (BSFC) point, (iii) engine off at stop, (iv) regeneration recovers 15–30 % of braking energy.
- Example: series path with η_{ICE}=0.35, gen 0.92, conv 0.96, motor 0.92, gear 0.97 gives 0.35×0.92×0.96×0.92×0.97 = **0.276 (27.6 %)**, while the direct parallel path gives 0.35×0.95 = 0.33 (33 %).

5. Configuration and Control of Motor Drives

(a) DC motor drives

Separately excited DC motor: $V_a = E + I_a R_a$, $E = K_e \varphi \omega$, $T = K_t \varphi I_a$. Speed is controlled by **armature voltage** (below base speed, constant torque) and by **field weakening** (above base speed, constant power).



- A DC-DC chopper (two-quadrant or four-quadrant H-bridge) supplies the armature from the battery. Control: inner current loop (fast) limits torque; outer speed loop gives reference current.
- + Simple control, good low speed torque; - brushes, commutator wear, low speed limit. Now largely replaced.

(b) Induction motor drives

- Battery -> **three-phase PWM inverter** -> IM. Configured for V/f control or **field-oriented control** with encoder /

sensorless speed estimation.

- FOC: i_{sd} (flux) and i_{sq} (torque) regulated by PI controllers in the rotating frame; field weakening above base speed using flux reference $\psi_r^* \propto 1/\omega$.

(c) Permanent magnet motor drives

- Inverter-fed PMSM with rotor position sensor (resolver). **Maximum torque per ampere (MTPA)** below base speed and **flux weakening** (negative i_d) above base speed. BLDC drive uses six-step commutation with Hall sensors.

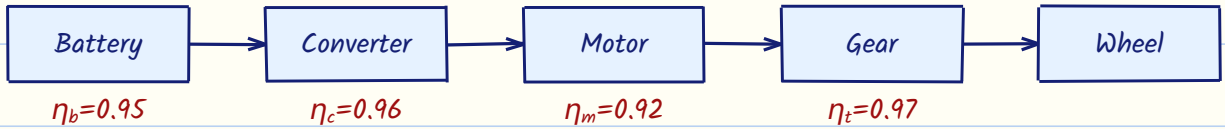
(d) Switched reluctance motor drives

- Asymmetric half-bridge converter per phase, rotor position feedback. Low speed: current (hysteresis) control; high speed: single-pulse angle control. Torque sharing function reduces ripple.

6. Drive System Efficiency

The **overall drive system efficiency** from battery to wheel is the product of the efficiencies of converter, motor and transmission. It is not constant - it depends on torque and speed (efficiency map).

Drive system efficiency = product of efficiencies in the chain



$\eta = 0.95 \times 0.96 \times 0.92 \times 0.97 = 0.81$ (81 %)

- Motor efficiency map: peak 90-96 % in middle speed and 50-80 % load; efficiency falls at very low torque (fixed losses) and at very high speed / torque (copper and iron losses).
- Losses: **copper loss** I^2R, **iron loss** (hysteresis + eddy), **mechanical** (friction, windage), **switching and conduction loss** in converter, gear loss.
- Improve efficiency by: better design (IPM), field weakening, high-efficiency SiC devices, optimal gear ratio so the motor operates in its sweet spot, using regeneration.

$\eta_{regen} = \eta_t \eta_m \eta_c \eta_b \approx 0.7 - 0.8$ (braking energy returned to battery

Important questions: (1) Explain history and social importance of HEV. (2) Series, parallel and series-parallel topologies with figures. (3) Power flow control in hybrid modes. (4) Control of DC / IM / PM / SRM drives. (5) Drive system efficiency.